CEUE203 - OOP25AIML034

PRACTICAL – 4

Aim:

1. Password Strength Checker: To develop a Java program using regular expressions and string methods to check password strength based on length, uppercase letters, digits, and special characters, and classify passwords as Weak, Medium, or Strong.

2. Chat-Log Filter: To develop a Java program that filters chat logs based on a user-provided keyword, skips malformed lines, performs case-insensitive searching, and displays the matching messages.

3. Template Filler (Advanced Learners): To develop a Java program using the Pattern and Matcher classes to identify and replace placeholders in a template with corresponding values, using [?] for missing values.

Code Screenshot:

```
class PasswordChecker {


   public static String strength(String pw) {

      int count = 0;



      if (pw.length() >= 8) count++;

      if (pw.matches(".*[A-Z].*")) count++;

      if (pw.matches(".*[0-9].*")) count++;

      if (pw.matches(".*[^a-zA-Z0-9].*")) count++;
```

```
        if (count <= 1) return "Weak";
        else if (count <= 3) return "Medium";
        else return "Strong";
    }

    public static void check(String pw) {
        System.out.println("Password: " + pw);
        System.out.println("Length >= 8: " + (pw.length() >= 8));
        System.out.println("Uppercase letter: " + pw.matches(".*[A-Z].*"));
        System.out.println("Contains digit: " + pw.matches(".*[0-9].*"));
        System.out.println("Special character: " +
                pw.matches(".*[^a-zA-Z0-9].*"));
        System.out.println("Strength: " + strength(pw));
        System.out.println();
    }
}

public class Main {
    public static void main(String[] args) {
        PasswordChecker.check("abc");
        PasswordChecker.check("Abcd1234!");
        PasswordChecker.check("hello123");
```

```
        PasswordChecker.check("Hello");
    }
}
```

2.

```
import java.util.Scanner;
import java.util.Locale;

class ChatFilter {

    public static void filter(String[] logs, String keyword) {
        StringBuilder report = new StringBuilder();
        int count = 0;

        for (String line : logs) {
            String[] parts = line.split(" ", 3);

            if (parts.length < 3) {
                continue;
            }

            String time = parts[0];
```

```
            String user = parts[1];
            String message = parts[2];

            if (message.toLowerCase(Locale.ROOT)
                    .contains(keyword.toLowerCase(Locale.ROOT))) {
                count++;
                report.append(time)
                        .append(" ")
                        .append(user)
                        .append(": ")
                        .append(message)
                        .append("\n");
            }
        }

        System.out.println("Matches: " + count);
        System.out.print(report);
    }
}

public class Main {
    public static void main(String[] args) {
```

```
        String[] logs = {
            "10:05 alice Hello there",
            "10:06 bob How are you?",
            "malformed",
            "10:07 riya Good morning"
        };

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter keyword: ");
        String keyword = sc.nextLine();

        ChatFilter.filter(logs, keyword);

        sc.close();
    }
}
```

3.

```
import java.util.regex.Matcher;
import java.util.regex.Pattern;
```

```
class TemplateFiller {


    public static String fillTemplate(
            String template, String[] names, String[] values) {


        Pattern pattern = Pattern.compile("\\{(\\w+)\\}");
        Matcher matcher = pattern.matcher(template);


        StringBuilder result = new StringBuilder();
        int lastEnd = 0;


        while (matcher.find()) {
            result.append(template, lastEnd, matcher.start());


            String placeholder = matcher.group(1);
            String replacement = "[?]";


            for (int i = 0; i < names.length; i++) {
                if (names[i].equals(placeholder)) {
                    replacement = values[i];
                    break;
                }
```

```
            }

            result.append(replacement);
            lastEnd = matcher.end();
        }

        result.append(template.substring(lastEnd));

        return result.toString();
    }
}

public class Main {
    public static void main(String[] args) {
        String template =
                "Dear {name}, order {id} ships {date}.";

        String[] names = {"name", "id"};
        String[] values = {"Riya", "A07"};

        String result = TemplateFiller.fillTemplate(
                template, names, values);
```

```
        System.out.println(result);
    }
}
```

Output Screenshot:

Output

```
Password: abc
Length >= 8: false
Uppercase letter: false
Contains digit: false
Special character: false
Strength: Weak

Password: Abcd1234!
Length >= 8: true
Uppercase letter: true
Contains digit: true
Special character: true
Strength: Strong

Password: hello123
Length >= 8: true
Uppercase letter: false
Contains digit: true
Special character: false
Strength: Medium

Password: Hello
Length >= 8: false
Uppercase letter: true
Contains digit: false
Special character: false
Strength: Weak
```

Output

```
Enter keyword: hello
Matches: 1
10:05 alice: Hello there
```

Output

```
Dear Riya, order A07 ships [?].
```